LDOS modulation $\delta\rho$ from the Born T-matrix sum $S(\mathbf{q})$

These notes follow the conventions of `QPI.pdf`: $S(\mathbf{q})$ is built from $g_0(\mathbf{k})\,T\,g_0(\mathbf{k}-\mathbf{q})$, and the FT-LDOS is $\rho(\mathbf{q}) = \sum_{\mathbf{r}} e^{-i\mathbf{q}\cdot\mathbf{r}}\rho(\mathbf{r})$. On the lattice, integrals $\int d^2k/(2\pi)^2$ become $\frac{1}{N}\sum_{\mathbf{k}}$, and the impurity potential is allowed to depend on momentum, $T \to T(\mathbf{k},\mathbf{k}')$.

1 Conventions

The lattice has N sites $\mathbf{r}$ and N momenta $\mathbf{k}$ in the Brillouin zone. Operators are related by

$$\Psi_{\mathbf{r}} = \frac{1}{\sqrt{N}}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}}\,\Psi_{\mathbf{k}}, \qquad \Psi_{\mathbf{k}} = \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c^{\dagger}_{-\mathbf{k}\downarrow} \end{pmatrix}. \tag{1}$$

Both Nambu components of $\Psi_{\mathbf{r}}$ carry the same phase, since $c^{\dagger}_{\mathbf{r}\downarrow} = N^{-1/2}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}} c^{\dagger}_{-\mathbf{k}\downarrow}$.

The Green's function is built from $\Psi_{\mathbf{r}}$ and $\Psi^{\dagger}_{\mathbf{r}'}$, e.g. $G^R(\mathbf{r},\mathbf{r}';t) = -i\theta(t)\langle\{\Psi_{\mathbf{r}}(t),\Psi^{\dagger}_{\mathbf{r}'}(0)\}\rangle$. The creation operator is the Hermitian conjugate of the same transform, $\Psi^{\dagger}_{\mathbf{r}'} = N^{-1/2}\sum_{\mathbf{k}'} e^{-i\mathbf{k}'\cdot\mathbf{r}'}\Psi^{\dagger}_{\mathbf{k}'}$, so the two ends carry opposite phases:

$$G(\mathbf{r},\mathbf{r}') = \frac{1}{N}\sum_{\mathbf{k},\mathbf{k}'} e^{i\mathbf{k}\cdot\mathbf{r}}\,G(\mathbf{k},\mathbf{k}')\,e^{-i\mathbf{k}'\cdot\mathbf{r}'}. \tag{2}$$

This is the change of basis $\langle\mathbf{r}|G|\mathbf{r}'\rangle = \sum\langle\mathbf{r}|\mathbf{k}\rangle\langle\mathbf{k}|G|\mathbf{k}'\rangle\langle\mathbf{k}'|\mathbf{r}'\rangle$ with $\langle\mathbf{r}|\mathbf{k}\rangle = e^{i\mathbf{k}\cdot\mathbf{r}}/\sqrt{N}$. As a check, a translation-invariant $G(\mathbf{k},\mathbf{k}') = \delta_{\mathbf{k}\mathbf{k}'}\,g_0(\mathbf{k})$ gives $G_0(\mathbf{r},\mathbf{r}') = \frac{1}{N}\sum_{\mathbf{k}} e^{i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}\,g_0(\mathbf{k})$, which depends only on $\mathbf{r}-\mathbf{r}'$ (cf. eq. (13) of `QPI.pdf`).

The bare retarded and advanced Green's functions are

$$g_0^R(\mathbf{k},\omega) = [\omega + i\gamma - H_{\mathrm{BdG}}(\mathbf{k})]^{-1}, \qquad g_0^A(\mathbf{k},\omega) = \left[g_0^R(\mathbf{k},\omega)\right]^{\dagger}, \tag{3}$$

and in general $G^A(\mathbf{r},\mathbf{r}') = [G^R(\mathbf{r}',\mathbf{r})]^{\dagger}$, so on the diagonal $G^A(\mathbf{r},\mathbf{r}) = [G^R(\mathbf{r},\mathbf{r})]^{\dagger}$.

2 Born approximation and $S(\mathbf{q})$

For an impurity $H_{\mathrm{imp}} = \sum_{\mathbf{k}\mathbf{k}'}\Psi^{\dagger}_{\mathbf{k}}T(\mathbf{k},\mathbf{k}')\Psi_{\mathbf{k}'}$ with $T(\mathbf{k},\mathbf{k}')^{\dagger} = T(\mathbf{k}',\mathbf{k})$, the first-order correction is

$$\delta G^R(\mathbf{k},\mathbf{k}') = g_0^R(\mathbf{k})\,T(\mathbf{k},\mathbf{k}')\,g_0^R(\mathbf{k}'). \tag{4}$$

Following eq. (21) of `QPI.pdf`, define

$$S(\mathbf{q}) = \sum_{\mathbf{k}} g_0^R(\mathbf{k})\,T(\mathbf{k},\mathbf{k}-\mathbf{q})\,g_0^R(\mathbf{k}-\mathbf{q}) = \sum_{\mathbf{k}}\delta G^R(\mathbf{k},\mathbf{k}-\mathbf{q}), \tag{5}$$

a 2×2 Nambu matrix. This is what the code computes (`G0k`, `T`, `G0kmq`). Both Green's functions in S are retarded.

3 Real-space diagonal

Setting $\mathbf{r}' = \mathbf{r}$ in (2) and writing $\mathbf{k}' = \mathbf{k} - \mathbf{q}$:

$$\delta G^R(\mathbf{r},\mathbf{r}) = \frac{1}{N}\sum_{\mathbf{k},\mathbf{q}} e^{i\mathbf{k}\cdot\mathbf{r}}\,\delta G^R(\mathbf{k},\mathbf{k}-\mathbf{q})\,e^{-i(\mathbf{k}-\mathbf{q})\cdot\mathbf{r}} = \frac{1}{N}\sum_{\mathbf{q}} e^{+i\mathbf{q}\cdot\mathbf{r}}\,S(\mathbf{q}), \tag{6}$$

so $S(\mathbf{q}) = \sum_{\mathbf{r}} e^{-i\mathbf{q}\cdot\mathbf{r}}\,\delta G^R(\mathbf{r},\mathbf{r})$ is the standard Fourier transform of $\delta G^R(\mathbf{r},\mathbf{r}) = S(\mathbf{r})$, as in eqs. (28)–(29) of `QPI.pdf`.

The advanced part is the Hermitian conjugate of (6):

$$\delta G^A(\mathbf{r},\mathbf{r}) = \big[\delta G^R(\mathbf{r},\mathbf{r})\big]^\dagger = \frac{1}{N}\sum_{\mathbf{q}} e^{-i\mathbf{q}\cdot\mathbf{r}}\,S(\mathbf{q})^\dagger = \frac{1}{N}\sum_{\mathbf{q}} e^{+i\mathbf{q}\cdot\mathbf{r}}\,S(-\mathbf{q})^\dagger. \tag{7}$$

$S(-\mathbf{q})^\dagger$ is the advanced counterpart of $S(\mathbf{q})$ with the same momentum structure. Using $T(\mathbf{k},\mathbf{k}')^\dagger = T(\mathbf{k}',\mathbf{k})$ and shifting $\mathbf{k} \to \mathbf{k}-\mathbf{q}$ (cf. eqs. (22)–(23) of `QPI.pdf`),

$$S(-\mathbf{q})^\dagger = \sum_{\mathbf{k}} g_0^A(\mathbf{k}+\mathbf{q})\,T(\mathbf{k}+\mathbf{q},\mathbf{k})\,g_0^A(\mathbf{k}) = \sum_{\mathbf{k}} g_0^A(\mathbf{k})\,T(\mathbf{k},\mathbf{k}-\mathbf{q})\,g_0^A(\mathbf{k}-\mathbf{q}) = \sum_{\mathbf{k}} \delta G^A(\mathbf{k},\mathbf{k}-\mathbf{q}). \tag{8}$$

4 FT-LDOS

The spin-up electron LDOS is

$$\delta\rho(\mathbf{r},\omega) = -\frac{1}{\pi}\,\mathrm{Im}\;\delta G^R_{00}(\mathbf{r},\mathbf{r},\omega) = -\frac{1}{2\pi i}\left[\delta G^R(\mathbf{r},\mathbf{r}) - \delta G^A(\mathbf{r},\mathbf{r})\right]_{00}. \tag{9}$$

Substituting (6) and (7) gives $\delta\rho(\mathbf{r}) = \frac{1}{N}\sum_{\mathbf{q}} e^{i\mathbf{q}\cdot\mathbf{r}}\,\delta\rho(\mathbf{q})$ with

$$\boxed{\delta\rho(\mathbf{q}) = \sum_{\mathbf{r}} e^{-i\mathbf{q}\cdot\mathbf{r}}\,\delta\rho(\mathbf{r}) = -\frac{1}{2\pi i}\left[S(\mathbf{q}) - S(-\mathbf{q})^\dagger\right]_{00},} \tag{10}$$

which is eq. (24) of `QPI.pdf`. The inverse transform $\delta\rho(\mathbf{r}) = \frac{1}{N_q^2}\sum_{\mathbf{q}} e^{+i\mathbf{q}\cdot\mathbf{r}}\,\delta\rho(\mathbf{q})$ is `numpy.fft.ifft2`, as used in the code.

Relation to the earlier $\mathbf{k}+\mathbf{q}$ form. An earlier version of the code used $S_+(\mathbf{q}) = \sum_{\mathbf{k}} \delta G^R(\mathbf{k},\mathbf{k}+\mathbf{q}) = S(-\mathbf{q})$. In terms of S_+, (10) reads $\delta\rho(\mathbf{q}) = \frac{i}{2\pi}\left[S_+(-\mathbf{q}) - S_+(\mathbf{q})^\dagger\right]_{00}$. That is the same expression with $\mathbf{q}$ relabelled, and it is the source of the apparent sign difference in $\mathbf{q}$. The two codes give bitwise-identical $\delta\rho$.

Retarded vs. advanced. Both terms in (10) contain one retarded and one advanced function at the same $\mathbf{q}$. Swapping $G^R \leftrightarrow G^A$ throughout flips the overall sign of $\delta\rho$, since $-\frac{1}{\pi}\,\mathrm{Im}\,G^A = +\frac{1}{\pi}\,\mathrm{Im}\,G^R$. It does not change the sign of $\mathbf{q}$.

5 Consequences for the symmetry analysis

Because $\delta\rho(\mathbf{r})$ is real, $\delta\rho(-\mathbf{q}) = \delta\rho(\mathbf{q})^*$; this is manifest in (10). Hence Re $\delta\rho(\mathbf{q})$ is even and Im $\delta\rho(\mathbf{q})$ is odd under $\mathbf{q} \to -\mathbf{q}$. In two dimensions inversion is the rotation $C_2 \in C_{6v}$, with $\chi(C_2) = +d_\nu$ for A_1, A_2, E_2 and $-d_\nu$ for B_1, B_2, E_1. So

$$\text{Re }\delta\rho(\mathbf{q}) \in A_1 \oplus A_2 \oplus E_2, \qquad \text{Im }\delta\rho(\mathbf{q}) \in B_1 \oplus B_2 \oplus E_1, \qquad \text{Im }\delta\rho(\mathbf{r}) = 0. \tag{11}$$

6 Implementation notes

- On the centred q-mesh, $\mathbf{q} = (m_1\mathbf{b}_1 + m_2\mathbf{b}_2)/N_q$ sits at index $(m_2 + N_q/2,\, m_1 + N_q/2)$. Mesh axis 0 steps along $\mathbf{b}_2$ and axis 1 along $\mathbf{b}_1$. $g_0(\mathbf{k}-\mathbf{q})$ is `np.roll` of $g_0(\mathbf{k})$ by $+(m_2, m_1)\, N_k/N_q$.
- $V(\mathbf{k}, \mathbf{k}-\mathbf{q})$ is evaluated with $\mathbf{k}$ and $\mathbf{k}-\mathbf{q}$ each folded into the Wigner–Seitz cell, which keeps V Hermitian and C_{6v}-covariant.
- $S(-\mathbf{q})$ does not need a separate calculation. $-\mathbf{q}$ sits at index $(N_q - i) \bmod N_q$ along each axis. Since g_0 is periodic in $\mathbf{k}$ and V uses folded momenta, $S(\mathbf{q}+\mathbf{G}) = S(\mathbf{q})$, so the reindexed array is exact. This matches a direct recalculation of $S(-\mathbf{q})$ to $\sim 10^{-16}$ relative.
- The overall factor $1/N$ is omitted in the code (it only sets the scale).

7 Numerical check

On a 12×12 lattice ($N_q = N_k = 12$, $d + id$ pairing, $\omega = 1.5\,\Delta_{\mathrm{gap}}$, $\gamma = 10^{-2}$), with a generic Hermitian $V(\mathbf{k}, \mathbf{k}')$ that has no C_{6v} symmetry:

- $\delta G^R_{00}(\mathbf{k}, \mathbf{k}')$ was built in the full $(\mathbf{k}, \mathbf{k}')$ basis, rotated to sites using $\langle \mathbf{r}|\mathbf{k}\rangle = e^{i\mathbf{k}\cdot\mathbf{r}}/\sqrt{N}$, and $-\frac{1}{\pi}\,\mathrm{Im}\;\delta G^R_{00}(\mathbf{r}, \mathbf{r})$ taken directly.
- The code's path ((5), (10), then `ifft2`) reproduces it with ratio 1 ± 10^{-12}, and its imaginary part is $\sim 10^{-16}$.
- $S(\mathbf{q})$ equals $\sum_{\mathbf{r}} e^{-i\mathbf{q}\cdot\mathbf{r}}\, \delta G^R_{00}(\mathbf{r}, \mathbf{r})$ to 2×10^{-15} relative.
- Replacing $g_0^R \to g_0^A$ in S gives exactly $-\delta\rho(\mathbf{r})$.